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#Roll no - 25

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
from scipy.stats import ttest_ind, f_oneway
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Upload Dataset
uploaded = files.upload()
file = list(uploaded.keys())[0]
df = pd.read_csv(file)

print("Dataset Loaded Successfully!")
print("Shape:", df.shape)

# -----
# Step 3: Descriptive Statistics
# -----

print("\n--- DESCRIPTIVE STATISTICS ---")
print(df["Age"].describe())

print("\nMean:", df["Age"].mean())
print("Median:", df["Age"].median())
print("Mode:", df["Age"].mode()[0])
print("Variance:", df["Age"].var())
print("Standard Deviation:", df["Age"].std())

# -----
# Step 4: Correlation Analysis
# -----

print("\n--- CORRELATION ---")
print(df[["Age", "Fare", "SibSp", "Parch"]].corr())

sns.heatmap(
    df[["Age", "Fare", "SibSp", "Parch"]].corr(),
    annot=True,
    cmap="coolwarm"
)
plt.title("Correlation Matrix")
plt.show()

# -----
# Step 5 & 6: Hypothesis Testing - T Test
# -----

survived_age = df[df["Survived"] == 1]["Age"].dropna()
not_survived_age = df[df["Survived"] == 0]["Age"].dropna()

t_stat, p_value = ttest_ind(
    survived_age,

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        not_survived_age
    )

print("\n--- INDEPENDENT T-TEST ---")
print("T-statistic:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Result: Significant difference")
else:
    print("Result: No significant difference")

# -----
# Step 7: One-Way ANOVA
# -----

class1 = df[df["Pclass"] == 1]["Fare"].dropna()
class2 = df[df["Pclass"] == 2]["Fare"].dropna()
class3 = df[df["Pclass"] == 3]["Fare"].dropna()

f_stat, p_value = f_oneway(
    class1,
    class2,
    class3
)

print("\n--- ONE-WAY ANOVA ---")
print("F-statistic:", f_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Result: Significant difference")
else:
    print("Result: No significant difference")

# -----
# Step 8: Linear Regression
# -----

data = df[["Age", "Fare"]].dropna()

X = data[["Age"]]
y = data["Fare"]

model = LinearRegression()
model.fit(X, y)

prediction = model.predict(X)

print("\n--- LINEAR REGRESSION ---")
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", r2_score(y, prediction))

plt.scatter(X, y)
plt.plot(X, prediction)
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Age vs Fare - Linear Regression")
plt.show()

```

Choose Files train.csv

train.csv(text/csv) - 61194 bytes, last modified: 9/13/2026 - 100% done

Saving train.csv to train.csv

Dataset Loaded Successfully!

Shape: (891, 12)

--- DESCRIPTIVE STATISTICS ---

count 714.000000

mean 29.699118

std 14.526497

min 0.420000

25% 20.125000

50% 28.000000

75% 38.000000

max 80.000000

Name: Age, dtype: float64

Mean: 29.69911764705882

Median: 28.0

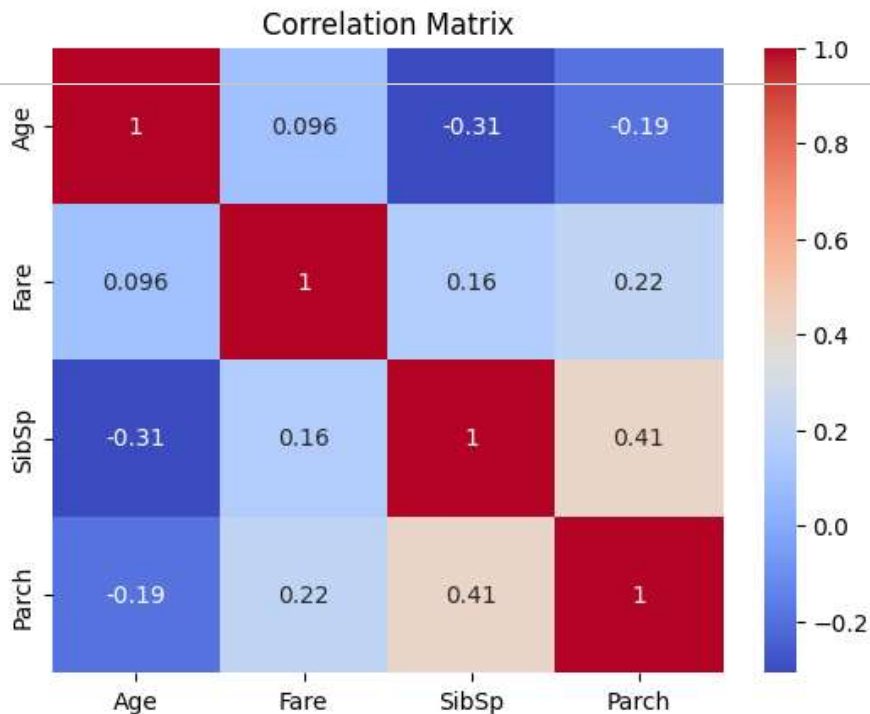
Mode: 24.0

Variance: 211.0191247463081

Standard Deviation: 14.526497332334044

--- CORRELATION ---

	Age	Fare	SibSp	Parch
Age	1.000000	0.096067	-0.308247	-0.189119
Fare	0.096067	1.000000	0.159651	0.216225
SibSp	-0.308247	0.159651	1.000000	0.414838
Parch	-0.189119	0.216225	0.414838	1.000000



--- INDEPENDENT T-TEST ---